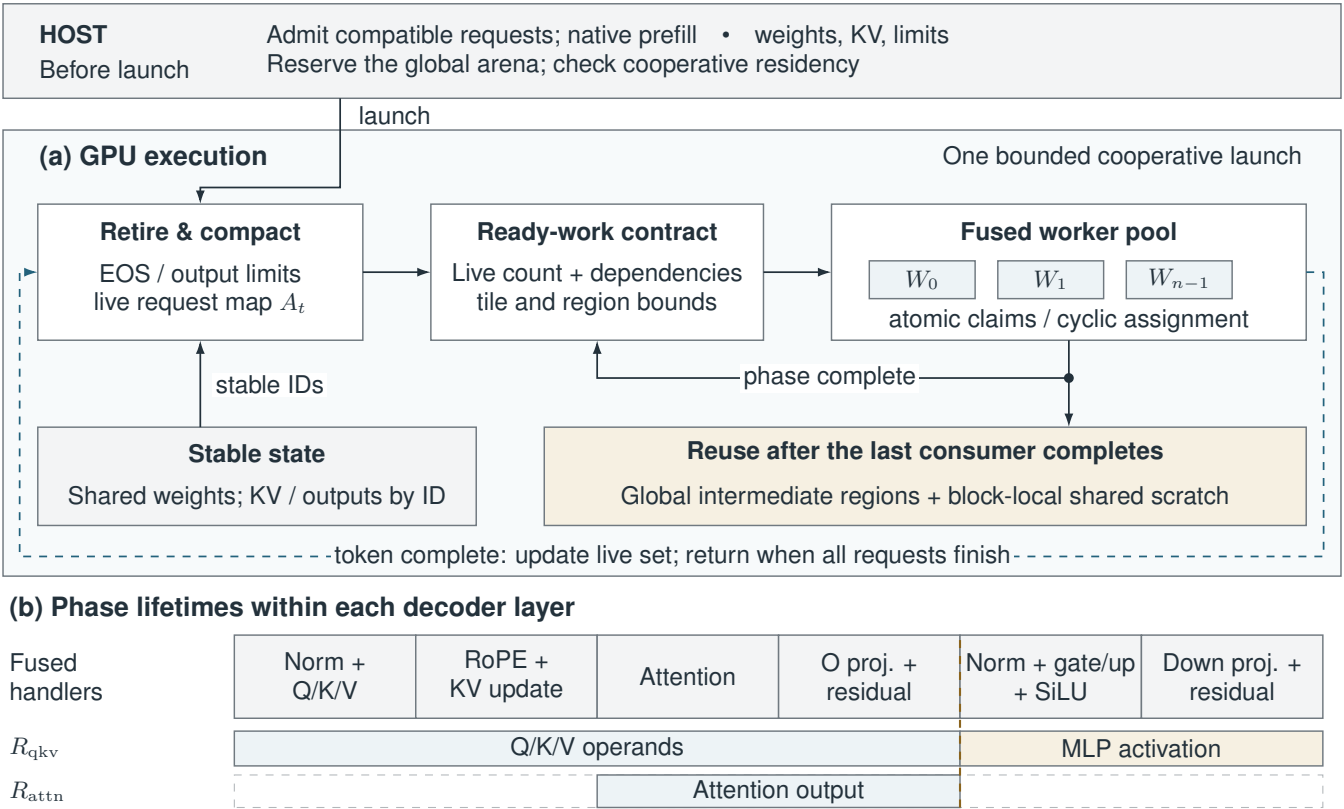

HOST
Before launch
Admit compatible requests; native prefill • weights, KV, limits
Reserve the global arena; check cooperative residency
launch
(a) GPU execution
One bounded cooperative launch
Retire & compact
EOS / output limits
live request map A_t
Ready-work contract
Live count + dependencies
tile and region bounds
Fused worker pool
W_0
W_1
W_{n-1}
atomic claims / cyclic assignment
stable IDs
phase complete
Stable state
Shared weights; KV / outputs by ID
Reuse after the last consumer completes
Global intermediate regions + block-local shared scratch
token complete: update live set; return when all requests finish
(b) Phase lifetimes within each decoder layer
Fused handlers
Norm + Q/K/V
RoPE + KV update
Attention
O proj. + residual
Norm + gate/up + SiLU
Down proj. + residual
R_{qkv}
Q/K/V operands
MLP activation
R_{attn}
Attention output